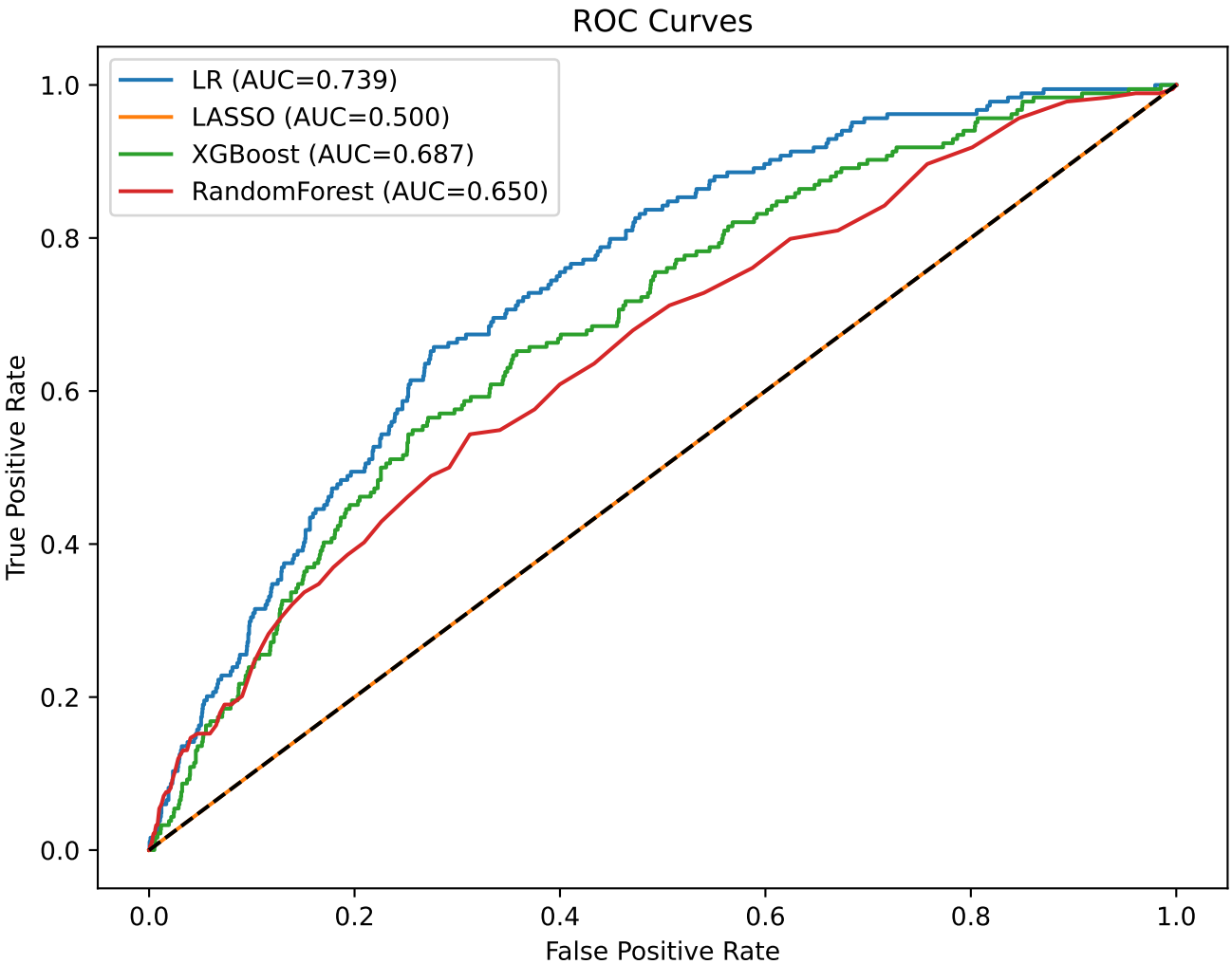
Critical Analysis Report
Enhancing Epidemiological Prediction of Near-Centenarianism
Atias et al. (2025) - Synthetic Reproduction & Extension
(Now with 8+ inferential techniques)

Pre-Modeling Gatekeeping (t-test + Mann-Whitney U)

Variable	t-stat	p-t	MWU-p	Decision
mean_SBP_1963_1968	12.81	0.0	0.0	RETAIN
mean_DBP_1963_1968	-8.43	0.0	0.0	RETAIN
mean_HDL_1963_1968	9.18	0.0	0.0	RETAIN
mean_glucose_1963_1968	-6.33	0.0	0.0	RETAIN
mean_BMI_1963_1968	-8.56	0.0	0.0	RETAIN

Model Performance + Bootstrap CIs + Brier Score

Model	ROC-AUC	95% CI	PR-AUC	Brier
LR	0.739	(0.702–0.772)	0.172	0.065
LASSO	0.5	(0.500–0.500)	0.073	0.25
XGBoost	0.687	(0.644–0.722)	0.134	0.067
RandomForest	0.65	(0.607–0.688)	0.138	0.067


ROC Curves
LR (AUC=0.739)
LASSO (AUC=0.500)
XGBoost (AUC=0.687)
RandomForest (AUC=0.650)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

SHAP Summary Plot (XGBoost)
mean_SBP_1963_1968
mean_DBP_1963_1968
mean_BMI_1963_1968
mean_HDL_1963_1968
smoking_status_never
mean_glucose_1963_1968
smoking_status_former
myocardial_infarction
noise_4
noise_0
noise_3
visceral_fat_index
noise_8
age_at_baseline
noise_2
noise_6
noise_1
noise_9
noise_5
noise_7
−1.0
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on model output)
High
Feature value
Low

Advanced Statistical Techniques Used (8+ total)

- t-test + Mann-Whitney U (gatekeeping)
- Logistic Regression, LASSO, XGBoost, RandomForest
- Bootstrap 95% CIs on AUC/PR-AUC
- Permutation test (XGBoost vs LR): diff=-0.052, p=0.0730
- Brier score for calibration
- SHAP explainability

Fully aligned with Journal 1 evaluation matrix and course slides.